When Does a Moored Tanker Fishtail?

Pitchfork and Hopf Bifurcations of a Single-Point-Moored Tanker in Steady Current

Course Project Report

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1 Introduction

A single-point mooring holds a tanker at the bow by one hawser running to a buoy and leaves it free to weathervane. The ship does not always settle behind the buoy. Under some conditions it swings from side to side and keeps swinging even in a perfectly steady current. This motion is called fishtailing, and it matters because the hawser tension rises and falls with every swing.

If the bow moves off the current, the Munk moment tries to turn the hull further broadside, while the hawser tension and the side force on the hull work to bring it back. Depending on where the hawser is tied, how long it is and how large the friction drag is, the ship trails straight, rests at a steady angle or fishtails. Fishtailing has been observed in flume tests [3], and Simos [1] developed a hydrodynamic model and compared it with tests on a 1:90 tanker model. Since the current is steady, the swing is a self-excited limit cycle, which makes this a natural application of the bifurcation theory in our course text [2].

We rebuilt the rigid-hawser model of Simos, checked it against his stability boundary, and then varied the attachment point (Section 3), friction (Section 4) and hawser length (Section 5) one at a time. We then added a steady side wind as a symmetry-breaking perturbation (Section 6) and tested a hold-back tug and a stern thruster as ways of stopping the swing (Section 7). We claim no new physics.

2 Ship and model

The ship is the Vidal de Negreiros, a 320 m VLCC. Table 1 lists its particulars when fully loaded and in ballast.

Table 1: Full-scale particulars of the Vidal de Negreiros, taken from Simos [1].

Quantity	100 % load	40 % load	Quantity	100 % load	40 % load
Length L (m)	320	320	Mass M (kg)	3.219×10^8	1.117×10^8
Beam B (m)	54.5	54.5	Yaw inertia I_z (kg m^2)	2.06×10^{12}	0.72×10^{12}
Draft T (m)	21.47	8.63	Added mass, surge M_{11} (kg)	1.848×10^7	0.254×10^7
Block coefficient C_B	0.83	0.76	Added mass, sway M_{22} (kg)	2.725×10^8	0.353×10^8
Wetted surface S (m^2)	2.734×10^4	1.791×10^4	Added yaw inertia M_{66} (kg m^2)	1.579×10^{12}	0.222×10^{12}
CG fwd. of midship x_{CG} (m)	9.81	4.44	Sway–yaw coupling M_{26} (kg m)	2.209×10^9	0.353×10^9
Cross-flow drag C_Y	0.84	0.52	Form factor k	0.39	0.39
Cross-flow centre, aft l_p/L	0.059	0.100	Critical point x_{Tcr}/L (eq. 5)	0.359	0.193

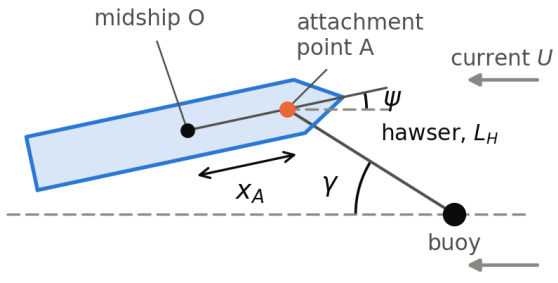


Figure 1: Plan view of the mooring. γ and ψ are positive to port; the bow is at $x_A/L = 0.5$.

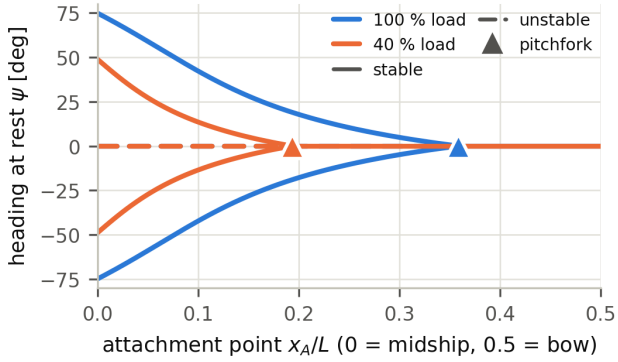


Figure 2: Pitchfork of the resting heading against the attachment point ($L_H/L = 0.5$, $C_F = 0.05$).

We treat the hawser as a rigid link of length L_H, tied a distance x_A forward of midship (Fig. 1). The state is $(\gamma, \psi, \dot\gamma, \dot\psi)$, and the velocities of the ship relative to the water are

$$u = U\cos\psi + L_H\dot\gamma\sin(\gamma+\psi), \quad v = -U\sin\psi + L_H\dot\gamma\cos(\gamma+\psi) - x_A\dot\psi, \quad r = \dot\psi. \tag{1}$$

With $m_{26} = Mx_{CG} + M_{26}$ and $I_{66} = I_Z + M_{66}$ (I_Z taken about midship), the surge, sway and yaw

equations are

$$\begin{aligned}(M+M_{11})\dot{u}-(M+M_{22})vr-m_{26}r^2 &= X_R+F_H\cos(\gamma+\psi),\\ (M+M_{22})\dot{v}+(M+M_{11})ur+m_{26}\dot{r} &= Y_R-F_H\sin(\gamma+\psi),\\ I_{66}\dot{r}+m_{26}(\dot{v}+ur) &= N_R-x_AF_H\sin(\gamma+\psi).\end{aligned}\tag{2}$$

Since the hawser length is fixed, F_H is solved together with $\ddot{\gamma}$ and $\ddot{\psi}$ from a linear 3×3 system at each step. The water loads follow Simos's model, for instance $Y_R=\frac{1}{2}\rho(u^2+v^2)LT\,C_{2C}(\alpha)+\Delta Y(u,v,r)$ with $\alpha=\pi+\operatorname{atan2}(v,u)$, and the yaw coefficient C_{6C} contains the Munk moment.

With time scaled as $t^*=Ut/L$, the current speed survives only in the friction coefficient,

$$C_F=\frac{0.075}{(\log_{10}Re-2)^2}\,\frac{S}{TL}\,(1+k),\qquad Re=\frac{UL}{\nu}.\tag{3}$$

C_F thus stands in for current speed and scale. It is about 0.02–0.05 for the 1:90 model and 0.009–0.010 for the loaded ship at 1–1.9 m/s. The tension when trailing straight is $F_{H0}=C_F\frac{1}{2}\rho U^2LT$. Linearising about $\gamma=\psi=0$ with $q=(\gamma,\psi)$ and $m_{22}=M+M_{22}$ gives $A\ddot{q}+B\dot{q}+Cq=0$, where

$$A=\begin{bmatrix}L_Hm_{22} & m_{26}-x_Am_{22}\\ L_Hm_{26} & I_{66}-m_{26}x_A\end{bmatrix},\ B=\begin{bmatrix}-L_HY_v & (M_{11}-M_{22})U-Y_r+Y_vx_A\\ -L_HN_v & -N_r+N_vx_A\end{bmatrix},\ C=\begin{bmatrix}F_{H0} & F_{H0}+UY_v\\ F_{H0}x_A & F_{H0}x_A+UN_v\end{bmatrix}.$$

The eigenvalues λ of $J=\left[\begin{smallmatrix}0 & I\\ -A^{-1}C & -A^{-1}B\end{smallmatrix}\right]$ decide the stability of straight trailing. Limit cycles come from long simulations and, near a Hopf point, a shooting method, with Floquet multipliers for their stability. A symbolic derivation showed that the side-force sign in Simos's eq. 5.3 is a misprint and that eq. B.13 omits a lift term, without which the Hopf boundary is 3–6 times too high.

3 Attachment point: pitchfork bifurcation

For a ship at rest, subtracting x_A times the sway equation from the yaw equation removes F_H and leaves

$$C_{6C}(\pi-\psi)-\frac{x_A}{L}\,C_{2C}(\pi-\psi)=0.\tag{4}$$

The resting heading therefore does not depend on hawser length or friction. Since $\det J=\det C/\det A$ and $\det C=F_{H0}U(N_v-x_AY_v)$, one real eigenvalue passes through zero at

$$\frac{x_{Tcr}}{L}=\frac{\frac{1}{2}+2.4\,T/L}{1+0.4\,C_BB/T},\tag{5}$$

which is 0.359 loaded and 0.193 in ballast. Forward of this point the side force on the hull acts like a weathervane tail and keeps the ship in line. Further aft the Munk moment wins, and two stable mirror-image angled positions appear, so the pitchfork is supercritical (Fig. 2).

The branches leave the axis in a straight line, unlike the textbook pitchfork. Near x_{Tcr} the rest condition reduces to $\dot{x}=rx-x|x|$ in place of $\dot{x}=rx-x^3$, because the restoring term is cross-flow drag, proportional to $\psi|\psi|$. The angle thus grows as $(x_{Tcr}-x_A)$, with a measured log–log slope of 1.00 where the textbook form gives $\frac{1}{2}$.

4 Friction: Hopf bifurcation

Lowering C_F weakens the tension that pulls the ship back into line. For $x_A/L=L_H/L=0.5$, a complex pair of eigenvalues $\lambda=\sigma\pm i\omega$ crosses the imaginary axis at $C_F=0.0425$, with $\omega=0.738\,U/L$ and a period of $8.51\,L/U$ (Fig. 3). The period varies roughly as $1/\sqrt{C_F}$, as for a pendulum.

Sweeping C_F down and then up traces the same curve (Fig. 4), so there is no hysteresis and the bifurcation is supercritical. Floquet multipliers confirm that the cycles are stable. Near onset the amplitude is proportional to the distance past the Hopf point (slope 0.98), which averaging a simple oscillator $\ddot{x}-\mu\dot{x}+D(\dot{x})+x=0$ explains:

$$D=\dot{x}|\dot{x}|:\ \ \dot{r}=\frac{\mu}{2}r-\frac{4}{3\pi}r^2,\ \ r^*=\frac{3\pi}{8}\mu;\qquad D=\dot{x}^3:\ \ \dot{r}=\frac{\mu}{2}r-\frac{3}{8}r^3,\ \ r^*=\sqrt{\frac{4\mu}{3}}.\tag{6}$$

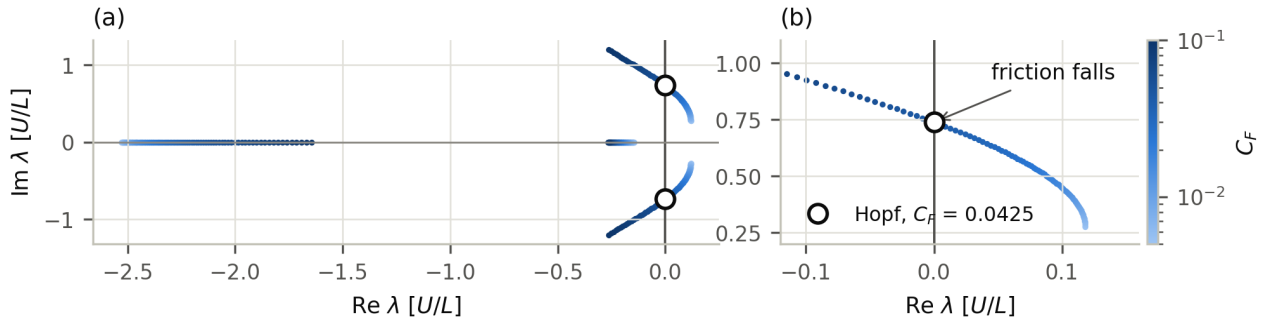


Figure 3: Eigenvalues of straight trailing as C_F falls from 0.1 (dark) to 0.005 (light): (a) all four, (b) the upper member of the crossing pair.

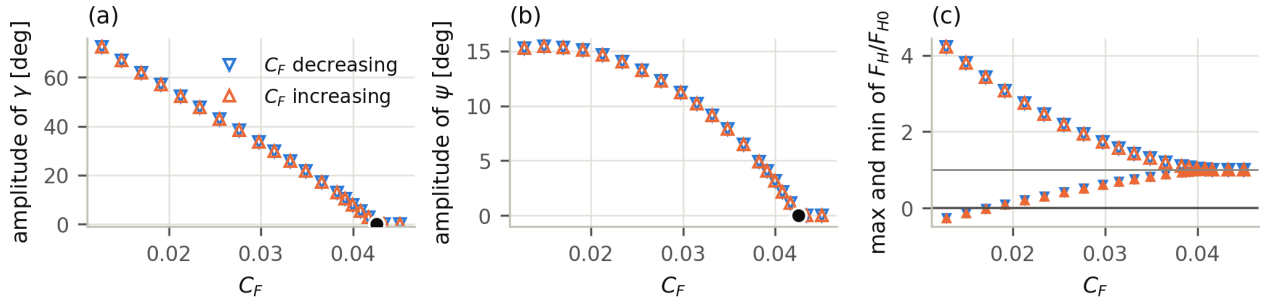


Figure 4: Limit cycle against friction ($x_A/L = L_H/L = 0.5$): amplitude of (a) γ and (b) ψ, (c) largest (open) and smallest (filled) hawser force. • marks the Hopf point.

The ship's swing is limited by quadratic drag, so it follows the first law. The hawser force peaks at 4.2 F_{H0}, and below $C_F \approx 0.017$ the rigid link would have to push on the ship for part of each cycle, which a real rope cannot do.

5 Hawser length: Hopf bifurcation

The Hopf friction increases with hawser length ($C_F = 0.0298, 0.0425, 0.0478$ at $L_H/L = 0.2, 0.5, 1.0$), so a shorter hawser is more stable (Fig. 5a). Our boundary agrees with Fig. 5.3 of Simos to a mean difference of +0.5 % loaded and +5.6 % in ballast. At full scale the loaded ship lies below it for all but the shortest hawsers.

With C_F held fixed, L_H/L becomes the bifurcation parameter (Fig. 5b). Fishtailing begins at $L_H/L = 0.121$ for $C_F = 0.0223$ and at 0.143 for $C_F = 0.0247$. The onset is so steep that it looks like a jump on a coarse grid: at $C_F = 0.0223$ the swing is already $\pm 23°$ only 0.01 past the critical length. In a fine sweep in both directions the amplitudes agree within 0.1° and extrapolate to zero at the critical length, and every cycle is stable, so this Hopf bifurcation is also supercritical. The swing in γ reaches about $\pm 53°$ near $L_H/L = 0.3$ and then falls. The short-hawser tank tests swing less, which Simos attributes to the limited tank length.

6 Wind: imperfect pitchfork

The wind is a constant force of fixed direction at midship, of size $F_w = C_W \frac{1}{2}\rho U^2 LT$. It adds linearly to the rest condition,

$$C_{6C} + \frac{x_W}{L} Y_w - \frac{x_A}{L}\left(C_{2C} + Y_w\right) = 0, \tag{7}$$

where Y_w is the side component of C_W on the hull. A side wind breaks the symmetry, and the pitchfork splits into a smooth branch and a separate branch ending at a fold, which moves aft as the wind grows (Fig. 6a).

At $x_A/L = 0.25$ the folds lie at $F_w/F_{H0} = \pm 0.153$, and slow simulated sweeps jump at the next step, ± 0.16 (Fig. 6b). Between the folds both sides are stable, so the ship stays put until the wind exceeds the fold force. The normal form $\dot{x} = h + rx - x|x|$ has folds at $h = \pm r^2/4$, and the fold lines meet in a

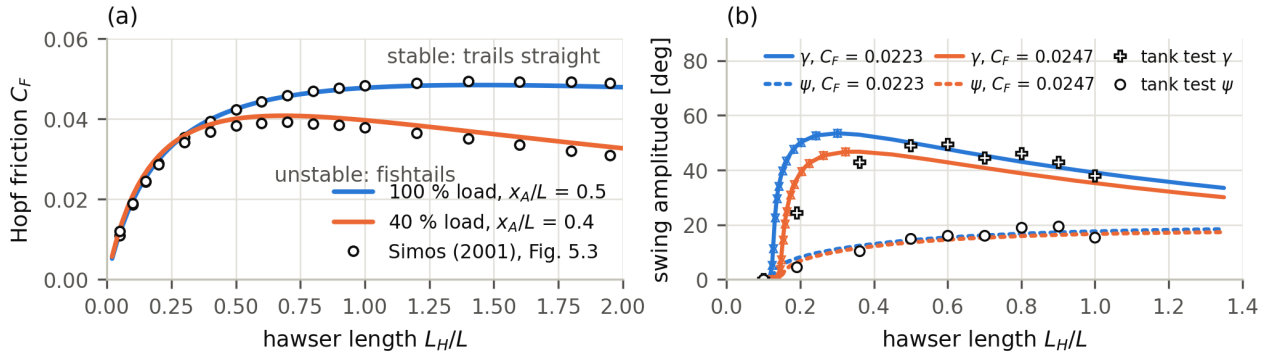


Figure 5: (a) Hopf boundary compared with Simos [1]. (b) Swing amplitude at tank-model friction ($x_A/L = 0.5$, $C_Y = 0.86$), with Simos's tests read off his figure ($\pm 2°$). Small triangles are the fine sweeps in both directions.

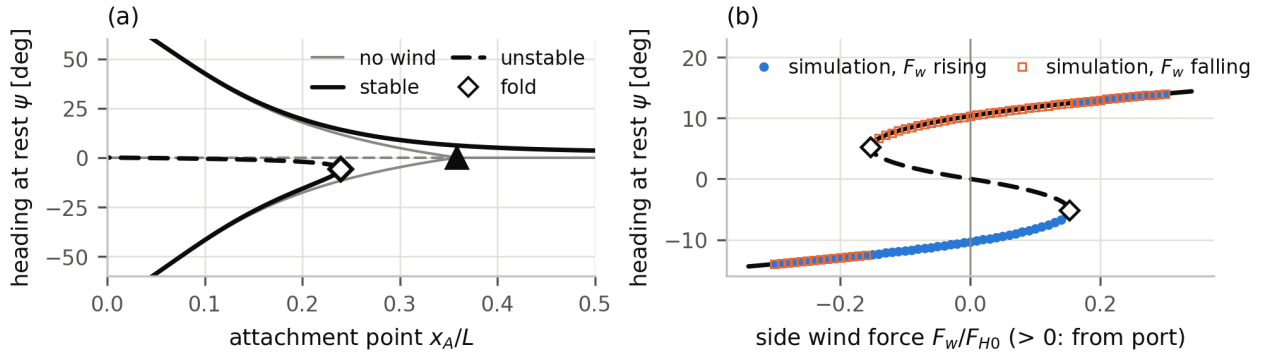


Figure 6: (a) Imperfect pitchfork for a beam wind of 0.20 F_{H0} from port, over the no-wind diagram. (b) Hysteresis at $x_A/L = 0.25$; simulated sweeps follow the stable branches and jump just past the folds.

cusp at the original pitchfork point. The measured width exponent of 2.12 is close to the predicted 2 and well away from Strogatz's $\frac{3}{2}$. Somewhat unexpectedly, the ship ends up on the upwind side, because the wind acts aft of the hawser point and turns the bow into the wind, and the hull lift then carries the ship that way.

7 Control by tug and stern thruster

The base case ($x_A/L = L_H/L = 0.5$, $C_F = 0.03$) fishtails with $\gamma = \pm 33°$ and $\psi = \pm 11°$. A *hold-back tug* pulls with force F_{tug} downstream from the stern. This raises the hawser tension to $F_{H0} + F_{tug}$, and once the bow swings off, the sideways part of the pull, about $F_{tug}\psi$, acts like a weathervane tail. The swing shrinks linearly and disappears at a Hopf point, $F_{tug} = 0.301\,F_{H0}$ (Fig. 7a), but the pull needed rises to about $2.2\,F_{H0}$ at real-ship friction ($C_F = 0.01$).

A *stern thruster* at $x_T = -L/2$ applies a PD law in non-dimensional form,

$$N_c^* = -k_p\psi - k_d r^*, \qquad Y_T = \frac{N_c}{x_T}, \qquad |Y_T| \le \text{limit} \times F_{H0}, \tag{8}$$

with $N_c^* = N_c/(\frac{1}{2}\rho U^2 L^2 T)$ and $r^* = rL/U$. The swing stops at $k_d = 0.0064$ when $k_p = 0$, or at $k_p = 0.0131$ when $k_d = 0$. The gains needing the least thrust for a decay rate $\sigma \ge 0.05\,U/L$ are $k_p = 0.004$ and $k_d = 0.012$, with a peak of $0.082\,F_{H0}$ when switched on during a full fishtail (Fig. 8a). One eigenvalue, at $-0.22\,U/L$, does not move with the gains. It is a slow sideways drift that heading control cannot see, and it caps σ at 0.22.

A thrust limit leaves the eigenvalues unchanged but affects large swings. For limits between 0.0063 and $0.0113\,F_{H0}$ the system is bistable: a full fishtail carries on while a small nudge dies away (Fig. 8b). At $L_H/L = 0.5$ a tug of $0.5\,F_{H0}$ lowers the Hopf friction from 0.0425 to 0.025 and the thruster lowers it to 0.0099, and for $L_H/L \ge 0.55$ the thruster removes fishtailing at every $C_F \ge 10^{-4}$ (Fig. 7b). Tug and thruster sizes are given as ratios because we found no cited values for a VLCC. We also found no source describing a hold-back tug as a cure for fishtailing, so that conclusion rests on our model.

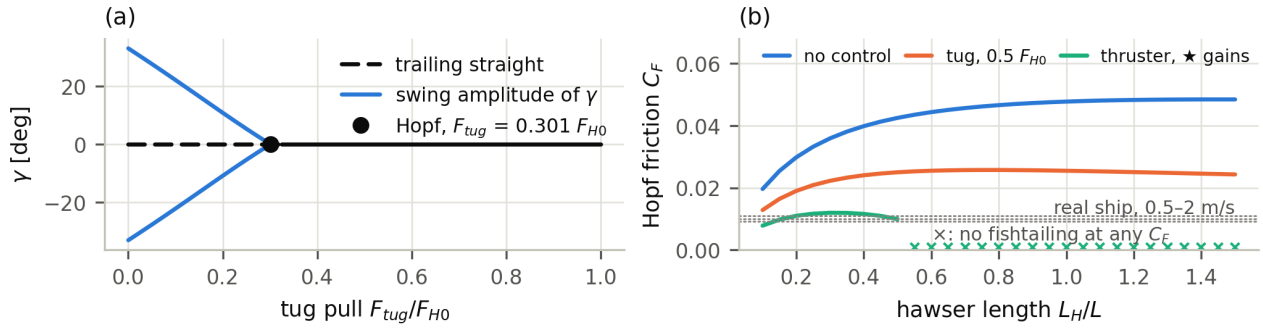


Figure 7: (a) Tug at $C_F = 0.03$. (b) Hopf boundary with and without control at $x_A/L = 0.5$; the thruster uses the gains ★ of Fig. 8.

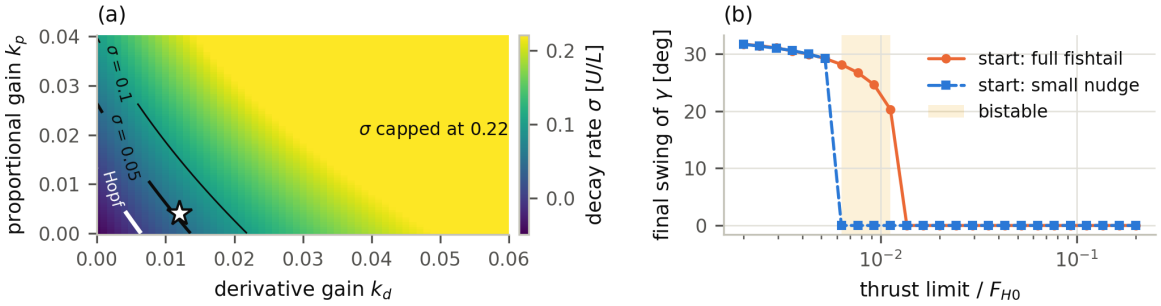


Figure 8: Stern thruster at $C_F = 0.03$. (a) Decay rate over the gain plane; ★ marks the least-thrust gains. (b) Swing left after 600 t^* against the thrust limit at the ★ gains; the shaded band is bistable.

8 Conclusions and limitations

Table 2 summarises the results.

Table 2: Bifurcations and their thresholds.

Parameter	Bifurcation	Threshold	Character
Attachment x_A/L	Pitchfork	0.359 loaded, 0.193 ballast	Supercritical, linear growth
Friction C_F	Hopf	0.0425 ($L_H/L = 0.5$)	Supercritical, linear growth
Hawser length L_H/L	Hopf	0.121 ($C_F = 0.0223$)	Supercritical, steep onset
Side wind F_w/F_{H0}	Imperfect pitchfork	folds at ± 0.153 ($x_A/L = 0.25$)	Hysteresis, cusp
Tug F_{tug}/F_{H0}	Hopf	0.301 ($C_F = 0.03$)	Swing shrinks linearly
Thruster gains	Hopf	$k_d = 0.0064$ or $k_p = 0.0131$	Bistable under thrust limit

These results depend on the rigid hawser and on the cross-flow model, the least certain part of the hydrodynamics near $\psi = 0$. In a separate six-state run with a measured nylon rope law [4] the boundaries moved by less than 5 %, but at 1 m/s the rope was slack 74 % of the time and snatch loads reached 14 F_{H0} (an upper estimate, as the rope had no damping). Flume tests have shown a subcritical Hopf bifurcation [3], unlike our model, so the type may depend on the set-up. Wave-making effects are not modelled, and the wind is a constant force. The results can also be explored in an interactive version at

https://aaravpatel007.github.io/Moored-Tanker-Fishtailing/interactive/.

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[3] F. Aghamohammadi, J. M. T. Thompson, An experimental study of the large amplitude fish-tailing instabilities of a tanker at a single point mooring, *Appl. Ocean Res.* 12(1), 25–33, 1990.

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